

BASIC LINUX AND VIM COMMAND

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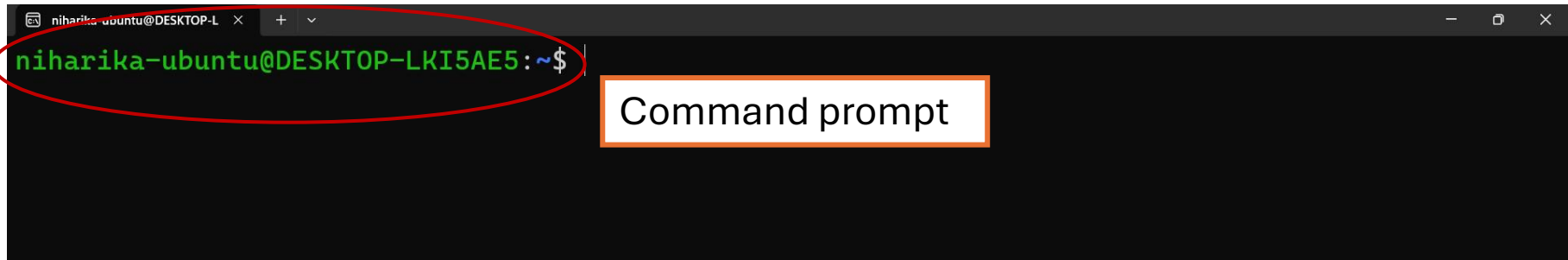
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India.



Terminal: Platform for working

Command to open a terminal: *Ctrl+Alt+T*

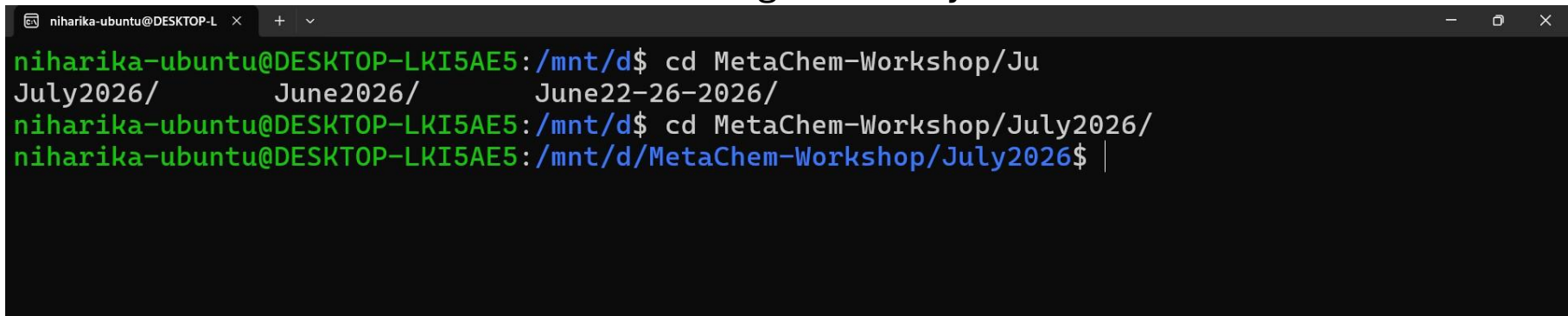


A terminal window with a dark background. The title bar shows 'niharika-ubuntu@DESKTOP-L'. The prompt is 'niharika-ubuntu@DESKTOP-LKI5AE5:~\$' in green text, which is circled in red. To the right of the terminal, there is a white box with an orange border containing the text 'Command prompt'.

```
niharika-ubuntu@DESKTOP-LKI5AE5:~$
```

Command prompt

cd : change directory

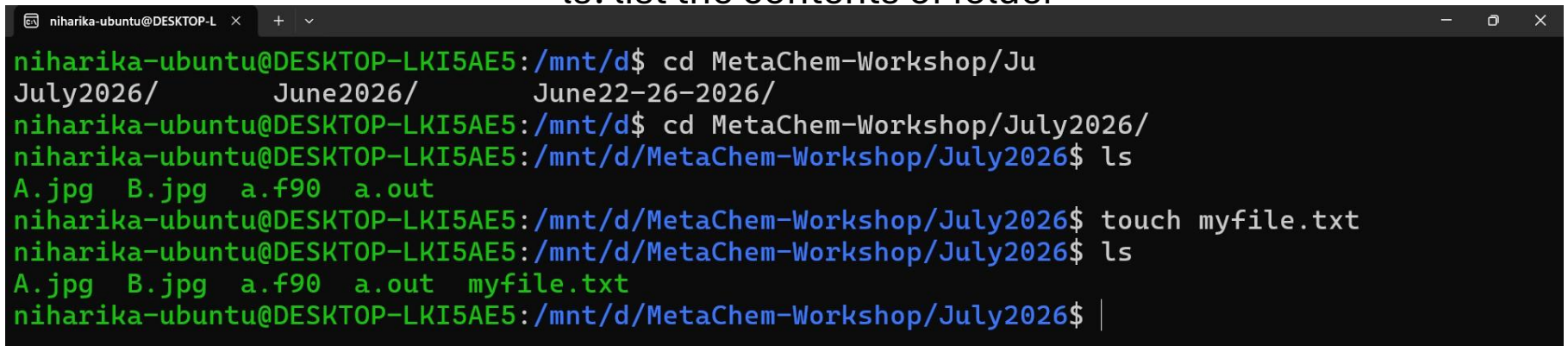


A terminal window showing directory navigation. The prompt is 'niharika-ubuntu@DESKTOP-LKI5AE5:/mnt/d\$'. The user enters 'cd MetaChem-Workshop/Ju' and 'July2026/'. The prompt changes to 'niharika-ubuntu@DESKTOP-LKI5AE5:/mnt/d\$'. The user enters 'cd MetaChem-Workshop/July2026/' and the prompt changes to 'niharika-ubuntu@DESKTOP-LKI5AE5:/mnt/d/MetaChem-Workshop/July2026\$'.

```
niharika-ubuntu@DESKTOP-LKI5AE5:/mnt/d$ cd MetaChem-Workshop/Ju
July2026/      June2026/      June22-26-2026/
niharika-ubuntu@DESKTOP-LKI5AE5:/mnt/d$ cd MetaChem-Workshop/July2026/
niharika-ubuntu@DESKTOP-LKI5AE5:/mnt/d/MetaChem-Workshop/July2026$ |
```

touch: create a empty file

ls: list the contents of folder



A terminal window showing file creation and listing. The prompt is 'niharika-ubuntu@DESKTOP-LKI5AE5:/mnt/d\$'. The user enters 'cd MetaChem-Workshop/Ju' and 'July2026/'. The prompt changes to 'niharika-ubuntu@DESKTOP-LKI5AE5:/mnt/d\$'. The user enters 'cd MetaChem-Workshop/July2026/' and the prompt changes to 'niharika-ubuntu@DESKTOP-LKI5AE5:/mnt/d/MetaChem-Workshop/July2026\$'. The user enters 'ls' and the output is 'A.jpg B.jpg a.f90 a.out'. The user enters 'touch myfile.txt' and the prompt changes to 'niharika-ubuntu@DESKTOP-LKI5AE5:/mnt/d/MetaChem-Workshop/July2026\$'. The user enters 'ls' and the output is 'A.jpg B.jpg a.f90 a.out myfile.txt'. The prompt changes to 'niharika-ubuntu@DESKTOP-LKI5AE5:/mnt/d/MetaChem-Workshop/July2026\$'.

```
niharika-ubuntu@DESKTOP-LKI5AE5:/mnt/d$ cd MetaChem-Workshop/Ju
July2026/      June2026/      June22-26-2026/
niharika-ubuntu@DESKTOP-LKI5AE5:/mnt/d$ cd MetaChem-Workshop/July2026/
niharika-ubuntu@DESKTOP-LKI5AE5:/mnt/d/MetaChem-Workshop/July2026$ ls
A.jpg B.jpg a.f90 a.out
niharika-ubuntu@DESKTOP-LKI5AE5:/mnt/d/MetaChem-Workshop/July2026$ touch myfile.txt
niharika-ubuntu@DESKTOP-LKI5AE5:/mnt/d/MetaChem-Workshop/July2026$ ls
A.jpg B.jpg a.f90 a.out myfile.txt
niharika-ubuntu@DESKTOP-LKI5AE5:/mnt/d/MetaChem-Workshop/July2026$ |
```

Command to open files in vim editor: *vi filename*

```
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d$ cd MetaChem-Workshop/Ju
July2026/      June2026/      June22-26-2026/
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d$ cd MetaChem-Workshop/July2026/
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026$ ls
A.jpg B.jpg a.f90 a.out
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026$ touch myfile.txt
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026$ ls
A.jpg B.jpg a.f90 a.out myfile.txt
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026$ vi myfile.txt |
```



The screenshot shows a terminal window with a dark background. The title bar at the top reads "niharika-ubuntu@DESKTOP-L" and includes standard window controls. On the left side, there is a vertical list of blue tilde (~) characters. The main area of the terminal is empty. At the bottom, the command `"myfile.txt" 0L, 0B` is entered, followed by the output `0,0-1` and `All`.

ls -ltr: to list the contents from old to new files along with date and time of the last update and file permissions (read-only, read and write, etc.)

ls -la : also list the hidden file in addition to other files.

Hidden files: some function files whose access is only to root-user.

```
niharika-ubuntu@DESKTOP-L x  Ubuntu
niharika-ubuntu@DESKTOP-LKI5AE5:/mnt/d/MetaChem-Workshop/July2026$ ls -ltr
total 744
-rwxrwxrwx 1 niharika-ubuntu niharika-ubuntu 106 Jul 25 05:37 a.f90
-rwxrwxrwx 1 niharika-ubuntu niharika-ubuntu 16264 Jul 25 05:37 a.out
-rwxrwxrwx 1 niharika-ubuntu niharika-ubuntu 27570 Jul 25 05:39 A.jpg
-rwxrwxrwx 1 niharika-ubuntu niharika-ubuntu 95751 Jul 25 05:41 B.jpg
-rwxrwxrwx 1 niharika-ubuntu niharika-ubuntu 196546 Jul 25 05:43 C.jpg
-rwxrwxrwx 1 niharika-ubuntu niharika-ubuntu 187116 Jul 25 05:44 D.jpg
-rwxrwxrwx 1 niharika-ubuntu niharika-ubuntu 50627 Jul 25 05:45 E.jpg
-rwxrwxrwx 1 niharika-ubuntu niharika-ubuntu 119231 Jul 25 05:53 F2.jpg
-rwxrwxrwx 1 niharika-ubuntu niharika-ubuntu 54203 Jul 25 05:54 F1.jpg
-rwxrwxrwx 1 niharika-ubuntu niharika-ubuntu 320 Jul 25 05:55 myfile.txt
niharika-ubuntu@DESKTOP-LKI5AE5:/mnt/d/MetaChem-Workshop/July2026$ ls -la
total 744
drwxrwxrwx 1 niharika-ubuntu niharika-ubuntu 4096 Jul 25 05:55 .
drwxrwxrwx 1 niharika-ubuntu niharika-ubuntu 4096 Jul 25 05:32 ..
-rwxrwxrwx 1 niharika-ubuntu niharika-ubuntu 27570 Jul 25 05:39 A.jpg
-rwxrwxrwx 1 niharika-ubuntu niharika-ubuntu 95751 Jul 25 05:41 B.jpg
-rwxrwxrwx 1 niharika-ubuntu niharika-ubuntu 196546 Jul 25 05:43 C.jpg
-rwxrwxrwx 1 niharika-ubuntu niharika-ubuntu 187116 Jul 25 05:44 D.jpg
-rwxrwxrwx 1 niharika-ubuntu niharika-ubuntu 50627 Jul 25 05:45 E.jpg
-rwxrwxrwx 1 niharika-ubuntu niharika-ubuntu 54203 Jul 25 05:54 F1.jpg
-rwxrwxrwx 1 niharika-ubuntu niharika-ubuntu 119231 Jul 25 05:53 F2.jpg
-rwxrwxrwx 1 niharika-ubuntu niharika-ubuntu 106 Jul 25 05:37 a.f90
-rwxrwxrwx 1 niharika-ubuntu niharika-ubuntu 16264 Jul 25 05:37 a.out
-rwxrwxrwx 1 niharika-ubuntu niharika-ubuntu 320 Jul 25 05:55 myfile.txt
niharika-ubuntu@DESKTOP-LKI5AE5:/mnt/d/MetaChem-Workshop/July2026$ |
```


mkdir: make a directory

cp : copy files to different locations

mv : move files to different locations. Also used for renaming files/folders

```
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026$ mkdir myfolder
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026$ cd myfolder/
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/myfolder$ cp ../myfile.txt .
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/myfolder$ ls
myfile.txt
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/myfolder$ |
```

```
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/myfolder$ ls ../
A.jpg C.jpg E.jpg F2.jpg H.jpg a.out myfolder
B.jpg D.jpg F1.jpg G.jpg a.f90 myfile.txt
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/myfolder$ mv ../a.f90 .
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/myfolder$ ls ../
A.jpg B.jpg C.jpg D.jpg E.jpg F1.jpg F2.jpg G.jpg H.jpg a.out myfile.txt myfolder
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/myfolder$ ls
a.f90 myfile.txt
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/myfolder$ |
```

mkdir: make a directory

cp : copy files to different locations

cp -r : copy folder to different locations

mv : move files to different locations. Also used for renaming files/folders

rm: remove/delete files

rm -r: remove/delete folders

rm -rf: remove/delete folders permanently

```
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026$ mkdir figures
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026$ mv *.jpg figures/
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026$
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026$
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026$ cp -r figures/ myfolder/
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026$ ls
a.out  figures  myfile.txt  myfolder
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026$ cd myfolder/
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/myfolder$ ls
a.f90  figures  myfile.txt
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/myfolder$ cd ../
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026$ rm -rf figures/
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026$ ls
a.out  myfile.txt  myfolder
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026$ rm a.out
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026$ ls
myfile.txt  myfolder
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026$ mv myfolder work-folder
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026$ ls
myfile.txt  work-folder
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026$ cd work-folder/
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$ ls
a.f90  figures  myfile.txt
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$ |
```

Accessing file contents without opening the file

Commands: *cat, grep, egrep, tail, head*

```
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$ ls
a.f90  figures  myfile.txt
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$ cat myfile.txt
# Input file for pw.x
&CONTROL
calculation = 'scf'
prefix = 'a'
outdir = '.'
/
&SYSTEM
ibrav = 1
celldm(1) = 7.50342
nat = 1
ntyp = 1
ecutwfc = 30
ecutrho = 220
/
&ELECTRONS
/
ATOMIC_SPECIES
Ni 58.6934 ni_pbe_v1.4.uspp.F.UPF
ATOMIC_POSITIONS (angstrom)
Ni 0.00000000 0.00000000 0.00000000
K_POINTS automatic
4 4 4 0 0 0
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$
```

```
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$ grep "ibrav" myfile.txt
ibrav = 1
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$ grep "ecut" myfile.txt
ecutwfc = 30
ecutrho = 220
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$ grep -A 2 "&SYSTEM" myfile.txt
&SYSTEM
ibrav = 1
celldm(1) = 7.50342
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$ grep -B 2 "&SYSTEM" myfile.txt
outdir = '.'
/
&SYSTEM
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$
```


Accessing file contents without opening the file

Commands: *cat*, *grep*, *egrep*, *tail*, *head*

```
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$ head myfile.txt
# Input file for pw.x
&CONTROL
calculation = 'scf'
prefix = 'a'
outdir = '.'
/
&SYSTEM
ibrav = 1
celldm(1) = 7.50342
nat = 1
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$ head -3 myfile.txt
# Input file for pw.x
&CONTROL
calculation = 'scf'
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$ tail myfile.txt
ecutrho = 220
/
&ELECTRONS
/
ATOMIC_SPECIES
Ni 58.6934 ni_pbe_v1.4.uspp.F.UPF
ATOMIC_POSITIONS (angstrom)
Ni 0.00000000 0.00000000 0.00000000
K_POINTS automatic
4 4 4 0 0 0
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$ tail -1 myfile.txt
4 4 4 0 0 0
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$ |
```

Accessing file contents without opening the file

Commands: *cat*, *grep*, *egrep*, *tail*, *head*

```
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$ grep -A 10 "&CONTROL" myfile.txt
t |tail -2
ntyp = 1
ecutwfc = 30
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$ grep -A 10 "&CONTROL" myfile.txt
t |head -4
&CONTROL
calculation = 'scf'
prefix = 'a'
outdir = '.'
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$ |
```

```
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$ egrep "nat|ntyp" myfile.txt
nat = 1
ntyp = 1
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$ grep "ecutrho" *txt
myfile-new.txt:ecutrho = 240
myfile-old.txt:ecutrho = 200
myfile.txt:ecutrho = 220
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$ ls *txt
myfile-new.txt  myfile-old.txt  myfile.txt
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$ ls
a.f90  figures  myfile-new.txt  myfile-old.txt  myfile.txt
niharika-ubuntu@DESKTOP-LKI5AE5: /mnt/d/MetaChem-Workshop/July2026/work-folder$ |
```

More functions for editing files in VIM

- ❑ Use arrow or “J K L H” keys to move the cursor up, down, left and right, respectively.
- ❑ Use “Y” key to copy the line. Use number keys to copy more than one line at a time. For eg: use “20y” to copy 20 lines from the line where the cursor is placed.
- ❑ Use “P” key to paste the copied contents in same or another file.
- ❑ Use “DD” key to delete the line. Use number keys to delete multiple lines. For eg: use “5dd” to delete 5 lines including the line where the cursor is placed.

More functions for editing files in VIM

- ❑ Use arrow or “J K L H” keys to move the cursor up, down, left and right, respectively.
- ❑ Use “Y” key to copy the line. Use number keys to copy more than one line at a time.
For eg: use “20y” to copy 20 lines from the line where the cursor is placed.
- ❑ Use “P” key to paste the copied contents in same or another file.
- ❑ Use “DD” key to delete the line. Use number keys to delete multiple lines. For eg: use “5dd” to delete 5 lines including the line where the cursor is placed.
- ❑ Use “R” for replacing one character at the position of the cursor.
- ❑ Use “Shift” + “R” keys for “REPLACE” mode.

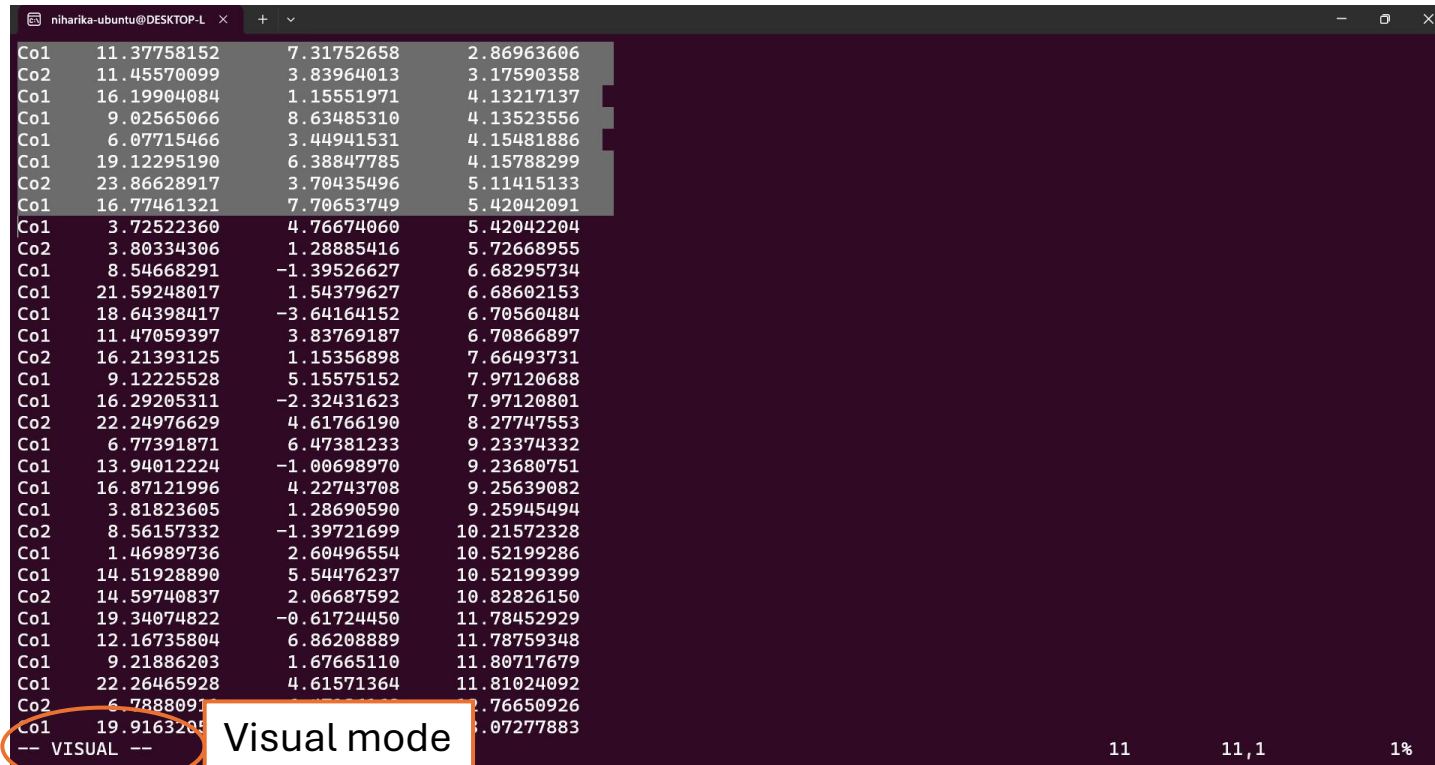
```
# Input file for pw.x
&CONTROL
calculation = 'scf'
prefix = 'a'
outdir = '.'
/
&SYSTEM
ibrav = 1
celldm(1) = 7.50342
nat = 1
ntyp = 1
ecutwfc = 30
ecutrho = 220
/
&ELECTRONS
/
ATOMIC_SPECIES
Ni 58.6934 ni_pbe_v1.4.uspp.F.UPF
ATOMIC_POSITIONS (angstrom)
Ni 0.0000000 0.0000000 0.0000000
K_POINTS automatic
4 4 4 0 0 0
```

```
# Input file for pw.x
&CONTROL
calculation = 'scf'
prefix = 'a'
outdir = '.'
/
&SYSTEM
ibrav = 1
celldm(1) = 6.43211
nat = 1
ntyp = 1
ecutwfc = 30
ecutrho = 220
/
&ELECTRONS
/
ATOMIC_SPECIES
Ni 58.6934 ni_pbe_v1.4.uspp.F.UPF
ATOMIC_POSITIONS (angstrom)
Ni 0.0000000 0.0000000 0.0000000
K_POINTS automatic
4 4 4 0 0 0

-- REPLACE --
```


More functions for editing files in VIM

- ❑ Use “Y” key to copy the line. Use number keys to copy more than one line at a time. For eg: use “20y” to copy 20 lines from the line where the cursor is placed.
- ❑ Use “P” key to paste the copied contents in same or another file.
- ❑ Use “DD” key to delete the line. Use number keys to delete multiple lines. For eg: use “5dd” to delete 5 lines including the line where the cursor is placed.



```
Co1 11.37758152 7.31752658 2.86963606
Co2 11.45570099 3.83964013 3.17590358
Co1 16.19904084 1.15551971 4.13217137
Co1 9.02565066 8.63485310 4.13523556
Co1 6.07715466 3.44941531 4.15481886
Co1 19.12295190 6.38847785 4.15788299
Co2 23.86628917 3.70435496 5.11415133
Co1 16.77461321 7.70653749 5.42042091
Co1 3.72522360 4.76674060 5.42042204
Co2 3.80334306 1.28885416 5.72668955
Co1 8.54668291 -1.39526627 6.68295734
Co1 21.59248017 1.54379627 6.68602153
Co1 18.64398417 -3.64164152 6.70560484
Co1 11.47059397 3.83769187 6.70866897
Co2 16.21393125 1.15356898 7.66493731
Co1 9.12225528 5.15575152 7.97120688
Co1 16.29205311 -2.32431623 7.97120801
Co2 22.24976629 4.61766190 8.27747553
Co1 6.77391871 6.47381233 9.23374332
Co1 13.94012224 -1.00698970 9.23680751
Co1 16.87121996 4.22743708 9.25639082
Co1 3.81823605 1.28690590 9.25945494
Co2 8.56157332 -1.39721699 10.21572328
Co1 1.46989736 2.60496554 10.52199286
Co1 14.51928890 5.54476237 10.52199399
Co2 14.59740837 2.06687592 10.82826150
Co1 19.34074822 -0.61724450 11.78452929
Co1 12.16735804 6.86208889 11.78759348
Co1 9.21886203 1.67665110 11.80717679
Co1 22.26465928 4.61571364 11.81024092
Co2 6.78880911 2.76650926 11.81024092
Co1 19.91632051 3.07277883 11.81024092
-- VISUAL --
```

- ❑ Use “V” key to go into “VISUAL” mode.
- ❑ Use arrow keys to select file contents.
- ❑ Use “Y” key to copy and “X” key to cut the selected file content.
- ❑ Use “P” key to paste the copied/cut content in the same or another file.
- ❑ Use “Esc” key to quit “VISUAL” mode.

THANK YOU!!!!
Keep practicing!!!